# A HYBRID INTERSTELLAR PROPULSION CONCEPT

*(QFHPS · Quantum-Fusion Hybrid Propulsion System)*

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*Technical study – version 5.0*

October 2026

*The concept was born during work on the hard-SF trilogy "L3"; this study presents it as a self-contained technical treatment.*

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*This propulsion concept comes from the science-fiction trilogy "L3" (The Trojan Observer · HORIZON · THE GARDENERS), available on Amazon. Share this file with anyone whose eyes it might open.*

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# A HYBRID INTERSTELLAR PROPULSION CONCEPT (QFHPS)

## The Quantum-Fusion Hybrid Propulsion System, with the construction specification of a reference interstellar ship

*Maciej Łuczewski*

*Technical study – version 5.0, October 2026*

## EDITORIAL NOTE

The present document is a **concept specification** of an interstellar drive and a reference ship, written in the convention of a scientific-technical study – with equations, with the status of every claim explicitly marked, and with a program of experimental verification. The author's intention is that it can be picked up by a physicist or an engineer and treated as a coherent, falsifiable engineering hypothesis. The concept was born during work on the hard-SF trilogy "L3"; this study presents it as a self-contained technical treatment, stripped of the literary layer.

Every claim and every parameter carries a **certainty status**:

**[F]** – *confirmed physics.* An element grounded in experimentally verified knowledge (as of 2025–2026).

**[E]** – *engineering extrapolation.* A conclusion derived from confirmed physics, but requiring an engineering breakthrough or scaling beyond the present state of the art.

This convention is a deliberate methodological choice: it allows the hard foundation to be separated from hypothesis, and it lets the reader see exactly where confirmed knowledge ends and extrapolation and speculation begin.

## ABSTRACT

The Quantum-Fusion Hybrid Propulsion System (QFHPS) is a concept for interstellar propulsion combining two complementary subsystems: the **Casimir Vacuum Modulator** – the main drive, drawing impulse from a controlled, asymmetric modulation of the zero-point energy of the electromagnetic field – and the **Magnetic Bussard Scoop** – a fusion reactor that gathers fuel (hydrogen, with an admixture of oxygen as catalyst) directly from the interstellar medium and the heliosphere. The first solves the problem of propulsion in the void (no reaction mass carried from Earth); the second, the problem of energy and fuel (replenished in flight). The architecture is deliberately redundant: the failure of one heart does not immobilize the ship.

The central physical thesis of this study is a reinterpretation of the Modulator's operation not as "thrust by recoil" but as a **controlled deformation of the spacetime metric in the immediate vicinity of the ship** – the ship is not *pushed by* space; it *falls down a gradient* that it itself creates ("Falling in Spacetime"). This framing resolves the apparent problem of momentum conservation (momentum is conserved in the system of ship + localized spacetime) and explains the observed properties of the drive: the absence of perceptible acceleration inside (the equivalence principle), the energy cost of braking maneuvers, and the gravitational "trace" left in spacetime.

The study further presents: a solution to the problem of heat rejection at the gigawatt scale through **neutrino cooling** (analogous to the cooling mechanisms of neutron stars), the specification of a fusion reactor with oxygen catalysis, a relativistic navigation architecture based on the aberration of starlight (SNAV/StarNAV), the full construction specification of the reference ship "Horizon", and a program of experimental verification whose first stage is a low-cost measurement of the net force from an asymmetric Casimir cavity.

# PART I. INTRODUCTION AND DESIGN PHILOSOPHY

## 1.1 The tyranny of the rocket equation

Classical reaction propulsion is governed by the Tsiolkovsky equation:

$$\Delta v = v_e \cdot \ln(m_0 / m_f), \qquad (1.1)$$

where $\Delta v$ is the velocity increment, $v_e$ the exhaust velocity, and $m_0/m_f$ the ratio of initial to final mass. **[F]** The logarithmic dependence is merciless: reaching speeds of the order of 0.1c with chemical propulsion ($v_e \approx 4.5$ km/s) would require a mass ratio exceeding the number of atoms in the observable Universe. Even a fusion drive with $v_e \approx 10{,}000$ km/s requires that the overwhelming majority of the initial mass be fuel – which *cannot be replenished anywhere* in interstellar space.

The conclusion is fundamental: interstellar flight within a reasonable time requires either (a) a drive that **does not carry reaction mass from Earth**, or (b) a drive that **refuels along the way**. The QFHPS implements *both* strategies at once – and that is the source of its name, "hybrid."

## 1.2 Two hearts

| Subsystem | Role | Solves the problem of |
|---|---|---|
| **Casimir Vacuum Modulator** | main drive (interstellar void) | reaction mass – draws impulse from the vacuum field, not from fuel |
| **Magnetic Bussard Scoop** | energy generator + maneuvering drive (heliosphere, star systems) | energy and fuel – gathers hydrogen from the medium |

The synergy is geometric and energetic at once: the Scoop works where there is matter (the dense solar wind, the circumstellar medium) and charges the energy storage; the Modulator carries the ship through the interstellar void, where the Scoop – for lack of raw material – "goes to sleep." Neither subsystem suffices alone; together they close the entire mission profile.

## 1.3 Design principle: existential redundancy

The supreme principle of the design is the rule that **no single failure may be a death sentence**. Loss of the Modulator means switching to thrust from the Scoop – slower, but sure (see emergency scenarios, Part IX). Every critical system has at least three independent implementations. In interstellar flight, where there is no servicing, no supply of spare parts, and no second chance, redundancy is not a luxury – it is a condition of survival.

# PART II. THE CASIMIR VACUUM MODULATOR – MAIN DRIVE

## 2.1 Zero-point energy – the foundation

Quantum mechanics forbids fields an exact zero. The harmonic oscillator of a field has a ground-state energy $E_0 = \frac{1}{2}\hbar\omega$; summing over all modes of the electromagnetic field we obtain the **zero-point energy (ZPE)** – a nonzero vacuum energy density even in the complete absence of particles. **[F]** The vacuum is not empty: it is a "boiling sea" of virtual particle–antiparticle pairs, constrained by the uncertainty relation:

$$\Delta E \cdot \Delta t \geq \hbar/2. \qquad (2.1)$$

The reality of this energy is not speculation – its direct, measurable manifestation is the Casimir effect.

## 2.2 The Casimir effect: static and dynamic

Two parallel conducting plates in vacuum restrict the set of allowed field modes between them. The vacuum energy density between the plates becomes *lower* than outside, which results in an attractive force **[F]**:

$$F_C / A = -\pi^2 \hbar c / (240\, d^4), \qquad (2.2)$$

where d is the plate separation and A their area. This force was measured by Lamoreaux (1997) and then with high precision by Mohideen and Roy (1998). The $1/d^4$ dependence is crucial and double-edged: it makes the effect **enormous at the nanometer scale and negligible macroscopically** – which is the central engineering challenge of the Modulator (see 2.4).

When the boundary conditions (plate separation) change **dynamically** at relativistic speed, the vacuum cannot relax in time and emits real photons out of the ZPE – this is the **dynamical Casimir effect (DCE)**, predicted by Moore (1970) and by Fulling and Davies (1976), and confirmed experimentally by Wilson et al. (2011) in a superconducting circuit with a modulated SQUID. **[F]** The DCE is physical proof that *modulating the boundaries of the vacuum allows real momentum and energy to be extracted from it* – this is the foundation on which the entire Modulator stands.

## 2.3 Asymmetry – from a conservative force to thrust

The static Casimir effect is symmetric and conservative – it yields no net propulsion. The Modulator introduces a **spacetime asymmetry**: the array of cavities is modulated so that the gradient of the vacuum energy density has a preferred direction. In a simplified treatment, the net force from a pair of cavities with different, time-varying separations $d_1(t)$, $d_2(t)$ is **[E]**:

$$F(t) = -(\hbar c \pi^2/240) \cdot A \cdot [\, 1/d_1(t)^4 - 1/d_2(t)^4 \,] \cdot \Psi_{coh}. \qquad (2.3)$$

The factor **$\Psi_{coh}$** (phase coherence) is the heart of the problem and has no full counterpart yet in the standard formalism – it describes the degree to which a million resonators pulse in a single, coherent phase. Without coherence, the contributions of the individual cavities average to zero; with full coherence they sum constructively, yielding macroscopic thrust. We mark $\Psi_{coh}$ as **[S]** – it is "an entanglement operator acting on the entire sphere of resonators," for which no rigorous theory exists.

## 2.4 The problem of scaling and coherence

The force from a single cavity is of the order of $10^{-12}$–$10^{-6}$ N. To obtain a useful thrust of the order of $10^6$ N (1000 kN), the contributions of **~$10^6$ resonators** pulsing synchronously at terahertz frequency (1–10 THz) must be summed **[E]**. This is a jump of ~16 orders of magnitude – comparable to the path "from a whisper to a detonation." Two sub-problems:

1. **Phase synchronization** of a million THz oscillators – requires a clock and diagnostics with attosecond resolution (solution: the plasma lens, Part VI).

2. **Power density and heat** – at 10 GW peak power, rejecting the heat with conventional radiators is impossible (solution: neutrino cooling, Part III).

## 2.5 THE SHIP THAT FALLS – metric modulation

The deepest and most important thesis of this annex concerns the *nature* of the motion induced by the Modulator. The naive picture ("the vacuum pushes the ship") immediately runs into the momentum-conservation objection: what does the reaction force act on? The answer requires abandoning the image of recoil in favor of the **geometry of spacetime**.

The Modulator, by producing a directional gradient of the vacuum energy density:

$$\nabla E_{ZPE} = (\hbar\omega/2) \cdot \nabla[\ \Sigma_k \sqrt{(k^2 + (mc/\hbar)^2)}\ ], \qquad (2.4)$$

locally modifies the spacetime metric in the immediate vicinity of the ship **[S]**:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}(x,t) \cdot f(E_{Casimir}), \qquad (2.5)$$

where $\eta_{\mu\nu}$ is the flat Minkowski metric and $h_{\mu\nu}$ a small perturbation controlled by the distribution of Casimir energy. The ship **does not move through space – it changes the shape of space around itself**. It creates a region of lowered energy density ahead of itself and of raised energy density behind: a spacetime "dip" tilted in the direction of flight. And – like a stone on a slope – **it falls into the dip it has carved for itself**.

This is the concept of **Falling in Spacetime**. Its consequences are precise and testable:

- **Conservation of momentum [E/S].** Momentum is not "created" and expelled outward. The ship and the curved spacetime surrounding it form a single dynamical system; momentum is conserved *within* that system. The ship falls not "into the vacuum," but into a local potential valley that it has itself produced. (The kinship with the Alcubierre metric (1994) is explicit here – with the difference that instead of exotic matter of negative energy density we use *controlled modulation of the vacuum*. Status of the full mechanism: **[S]**.)

- **Exotic matter – the objection reversed [F/S].** We turn the exotic-matter objection around: the Casimir vacuum **is** the only laboratory-known state with negative energy density relative to the background ($\rho = -\pi^2\hbar c/720 \cdot d^4$; it violates the weak energy condition – this is textbook physics, not a hypothesis) **[F]**. The Modulator does not postulate new matter – it scales up the only "negative" state the Universe has so far allowed us to build. The scaling remains **[S]**; the existence of the raw material is **[F]**.

- **The equivalence principle [F → E].** Inside a ship moving with constant gravitational acceleration it is impossible – locally – to distinguish that acceleration from a uniform gravitational field (Einstein, 1907). The crew of a "falling" ship is in a state of **free fall**: they feel no g-loads, just as astronauts in orbit do not. This is not a measurement defect – it is a direct consequence of the fact that the drive acts on *geometry*, not through *a force applied to the hull*.

- **The energy cost of braking [E].** "Falling" is a spontaneous, entropically favored tendency of a system to reach a state of lower energy – it is "free" in the sense in which coasting down a slope is free. But **stopping or turning back requires building a "rise" in front of oneself** – reversing the gradient – which is costly. This naturally explains why, in the mission profile, the braking phase is energetically the most expensive.

- **The trace in spacetime [S].** The ship's passage leaves a transient deformation of the metric – a "rut" in spacetime, decaying after the passage but briefly measurable. Such a signature – non-gravitational acceleration without mass ejection – is one of the observational predictions of the concept (Part XI.3).

- **Thermodynamics [E].** The drive converts ordered energy from the storage units (SMES) into disorder (DCE photons, heat), but in a way that *carves a useful gradient path*. The ship "rides the current of entropy" – which ties it together with neutrino cooling (Part III) into a single, thermodynamically consistent picture.

## 2.6 Modulator specification [E/S]

| Parameter | Value | Status |
|---|---|---|
| Location | ship's center of mass (sphere) | – |
| Number of micro-resonators | 1,048,576 (2^20), 16 tower-segments | [E] |
| Dimensions of a single resonator | 10 × 10 × 1 mm | [E] |
| Cavity separation | 10–100 nm, dynamic | [E] |
| Material | functionalized graphene + metamaterials | [E] |
| Modulation frequency | 1–10 THz | [E] |
| Peak power | ~10 GW | [E] |
| Maximum thrust | ~1000 kN (≈100 tonnes-force) | [S] |
| Coherence Ψ_coh | necessary condition, no full theory | [S] |

*Note on power: a photon drive with 1000 kN of thrust would require $P = F \cdot c \approx 3 \times 10^{14}$ W. The Modulator does not expel a momentum flux – the 10 GW covers the cost of maintaining the metric gradient (coherence dissipation, DCE losses, cooling), not the cost of "momentum fuel." The power–thrust relation is not subject to the photon limit; its exact form shares the [S] status of the h_μν mechanism.*

# PART III. NEUTRINO COOLING – HOW TO REJECT THE HEAT OF THE VACUUM

## 3.1 The problem: the radiative wall

A Modulator operating at a peak power of ~10 GW generates a stream of waste heat that cannot be rejected conventionally. The thermal radiation of a radiator is governed by the Stefan–Boltzmann law **[F]**:

$$P / A = \varepsilon \sigma T^4, \qquad (3.1)$$

where $\sigma = 5.67 \cdot 10^{-8}$ W·m$^{-2}$·K$^{-4}$. Even a radiator operating at the high temperature of 1500 K radiates on the order of 290 kW/m$^2$; rejecting 10 GW would require **tens of thousands of square meters** of radiator area – the mass and vulnerability to damage make this solution impossible for a ship of 10,000 tonnes. This is the "radiative wall": the fundamental barrier against which most high-energy propulsion concepts shatter.

## 3.2 The astrophysical precedent – how stars cool

Nature solved this problem long ago. Neutron stars, stars in the late phases of evolution, and matter on the eve of a supernova explosion **do not cool mainly by radiation from the surface – they cool by neutrino emission from the entire volume [F]**. Neutrinos, interacting only weakly (cross section of the order of $10^{-44}$ cm$^2$ at MeV energies), leave the interior of a star nearly unimpeded, carrying energy and entropy directly from the core into space. The channels include:

- **URCA processes** (direct and modified): alternating β decay and electron capture, $n \rightarrow p + e^- + \bar{\nu}_e$ and $p + e^- \rightarrow n + \nu_e$, in which every cycle carries out a neutrino–antineutrino pair;

- **pair annihilation** $e^+ + e^- \rightarrow \nu + \bar{\nu}$, as well as photoneutrino and plasmon processes (plasmon decay $\gamma^* \rightarrow \nu + \bar{\nu}$) in hot, dense plasma.

The key property: **neutrinos need no radiator surface**. They carry energy away from the volume, "transparent" to matter. This is exactly the property the Modulator needs.

## 3.3 The mechanism in the Modulator [E/S]

In the core of the Modulator, in the region of highest field-energy density and extreme ZPE modulation, part of the waste energy is channeled **not into phonon heat (which would have to be radiated away), but into the production of neutrino–antineutrino pairs** through coupling to the hot, dense resonator plasma (an analog of plasmon processes and pair annihilation). The neutrino flux leaves the ship isotropically and without interaction, carrying the excess entropy straight into space – **bypassing the radiative wall**.

Status: the mechanism is **[S]** (coupling of the modulated vacuum to the neutrino channel is not achievable in laboratory conditions today), but its *principle* is **[F]** – it is the same channel by which every neutron star cools. Mastery of neutrino cooling would be the breakthrough moving the Modulator from the realm of engineering impossibility into the realm of feasibility – and the precondition for raising the system's power beyond the radiative barrier.

## 3.4 Thermodynamic consistency

Neutrino cooling closes the picture of "the ship that falls": the drive converts ordered energy (SMES) into disorder, but **exports that disorder through the cleanest channel possible** – a stream of particles that do not come back and heat nothing along the way. The ship does not merely "ride the current of entropy" – it **exhales entropy in the form of neutrinos**, just as a star does. This makes the heat balance closed and physically consistent.

# PART IV. THE MAGNETIC BUSSARD SCOOP – THE SECOND HEART

## 4.1 The principle: a filling station that flies with you

The Bussard Scoop (concept: Robert W. Bussard, 1960) solves the fuel problem by collecting the diffuse hydrogen of the interstellar medium and the solar wind with a vast **magnetic field funnel**, compressing it and directing it into a fusion reactor **[E]**. In the QFHPS the Scoop plays two roles: it is the **energy generator** (charging the storage units that power the Modulator) and the **maneuvering drive** in matter-rich regions (the heliosphere, circumstellar systems).

The decades-old objection to the Bussard ramjet – that at relativistic speeds the drag of the magnetic funnel exceeds the thrust obtained – we accept in full and put to use **[F/E]**: the Scoop gathers fuel where things are slow and dense (perihelion, circumstellar systems); in interstellar

flight it works as a generator; and during braking its drag becomes a **magnetic brake** (cf. the magnetic sail brake of Zubrin–Andrews, 1991). The Scoop's thirty-percent share in deceleration (Table 5.1) is not reverse thrust – it is a funnel deliberately spread wide.

## 4.2 Plasma and fusion physics

The behavior of the collected hydrogen is described by the Vlasov–Maxwell equation for collisionless plasma; the characteristic scales are the Debye length and the plasma frequency **[F]**:

$$\lambda_D = \sqrt{(\varepsilon_0 k_B T / n e^2)}, \quad \omega_p = \sqrt{(n e^2 / m \varepsilon_0)}. \qquad (4.1)$$

The condition for fusion ignition is the Lawson criterion **[F]**:

$$n \cdot \tau_E > \sim 10^{21}\ m^{-3} \cdot s \cdot keV, \qquad (4.2)$$

and the basic energy channel is the proton–proton chain:

$$4\ {}^{1}H \rightarrow {}^{4}He + 2\ e^{+} + 2\ \nu_e + 26.7\ MeV. \qquad (4.3)$$

## 4.3 Oxygen as catalyst – the strategic role of Callisto

The p–p chain is slow – in the Sun a single proton waits billions of years for fusion, because the first step ($p + p \rightarrow d + e^{+} + \nu_e$) proceeds through the weak interaction and **no catalyst can accelerate it**. The reactor cannot wait – so it does not perform that step at all: **it bypasses it**, burning hydrogen in **the hot CNO cycle**, in which nuclei of carbon, nitrogen, and **oxygen** act as *catalysts* (they are not consumed, but repeatedly carry out the synthesis of helium from hydrogen via rapid proton captures) **[F]**. The precedent is astrophysical and hard: this is exactly how **classical novae** burn hydrogen – and their temperature regime ($1–4\times10^{8}$ K) overlaps the reactor's operating range (10–50 keV, Table 4.4). The bottleneck of the hot CNO cycle is the $\beta^{+}$ decays of the catalysts ($^{13}N$: 10 min; $^{15}O$: 122 s): each catalyst nucleus "processes" one helium nucleus per dozen-or-so minutes, so **the reactor's power scales with the mass of catalyst in the plasma**.

The reactor needs tonnes of oxygen, not a pinch – and hauling such masses out of Earth's gravity well is energetically irrational. The natural solution is to take on the catalyst along the way. **Callisto** – a moon of Jupiter whose surface is rich in water ice with oxidized admixtures, orbiting outside the planet's main radiation belt – is an example of a body that, after prior preparation of extraction infrastructure, could serve as the last supply point before leaving the Solar System: humanity's last filling station. **[E]**

## 4.4 Reactor, confinement, and energy conversion

| Parameter | Value | Status |
|---|---|---|
| Reactor type | spherical tokamak | [E] |
| Gain factor | Q > 50 | [E] |
| Plasma temperature | ~100 million K (10–50 keV) | [F/E] |
| Superconducting coils | 48 main + 128 corrective | [E] |
| Magnetic field | 20 T (center) → 0.001 T (funnel edge) | [E] |
| Collection funnel range | adjustable 100–10,000 km | [E] |
| Fusion power | 100 MW – 10 GW (per medium density) | [E] |
| Energy conversion | direct, magnetohydrodynamic (MHD), efficiency > 70% | [E] |

Plasma confinement is governed by magnetohydrodynamics; the important parameters are the plasma beta $\beta = p/(B^2/2\mu_0)$ and the magnetic Reynolds number $R_m = \mu_0 \sigma v L$ **[F]**. The heat of the "dirty" first wall is carried away by rings of silicon carbide and tungsten, cooled with liquid lithium (>1 $MW/m^2$) **[E]**.

# PART V. SYNERGY, FUEL CYCLE, AND OPERATING MODES

## 5.1 Operating modes (thrust profile)

| Phase | Modulator | Scoop | Acceleration |
|---|---|---|---|
| Launch / boost | ~90% | ~10% | 1 g |
| Interstellar cruise | ~100% | dormant | 0 (free flight at ~10% c) |
| Braking | ~70% | ~30% | 1 g (costly – gradient reversal) |

## 5.2 The fuel–energy cycle

The Scoop charges the energy storage (SMES 100 GWh + graphene ultracapacitors 10 GWh), from which the Modulator draws during cruise, when the Scoop – for lack of dense matter – is inactive. The key maneuver of the mission is **refueling at the Sun** (perihelion ~0.1 AU): in the dense solar wind the Scoop works at maximum efficiency, stockpiling fuel and energy for decades of flight: a clean flight through the void – falling in spacetime powered by energy of the Sun gathered in advance.

# PART VI. BREAKTHROUGH TECHNOLOGIES 2025–2026

Two discoveries from 2025–2026 move the hardest subsystems of the QFHPS from the category of "breakthrough challenges" to the category of "engineering problems."

## 6.1 The plasma lens for attosecond pulses

*Max Born Institute (Berlin) and DESY (Hamburg); Nature Photonics, November 2025.* **[F]**

The first active lens formed from plasma itself (a plasma channel in a hydrogen capillary), capable of focusing attosecond pulses with >80% transmission and an adjustable focal length (via plasma density). Two applications in the QFHPS:

- **The Bussard Scoop – sculpting the plasma "from within."** Instead of only squeezing the plasma with magnets from outside, the plasma lens allows (a) focusing the hydrogen stream into the reactor core with surgical precision (raising Q), (b) creating an additional, flexible confinement "wall" in the tokamak, (c) dynamically adjusting compression according to the density of the collected solar wind. **[E]**

- **The Casimir Modulator – attosecond diagnostics.** Attosecond-scale microscopy provides a tool for calibrating the THz resonators and **detecting phase errors** before they disturb the coherence Ψ_coh of a million cavities. These are the Modulator's missing "eyes." **[E]**

## 6.2 Detection of the Unruh effect via superradiance

*Physical Review Letters, 2024 (DOI 10.1103/PhysRevLett.133.200401).* **[F]**

The method converts the faint quantum signal of the vacuum (the Unruh effect / DCE) into a clear, amplified **superradiant flash** – excited atoms between mirrors of finesse $>10^6$ in a sub-resonant cavity. For the QFHPS this is the laboratory footbridge between "the Casimir effect on an optical bench" and vacuum modulation at drive scale: a tool for measuring and tuning the Modulator's coupling to the vacuum. **[E/S]**

# PART VII. THE REFERENCE SHIP "HORIZON" – CONSTRUCTION SPECIFICATION

To close the concept, we present a reference ship design – in the convention of engineering studies such as Project Daedalus (British Interplanetary Society, 1978) or Project Icarus: the specification illustrates the scale and systemic coherence of the solution; it does not prejudge the final construction.

## 7.1 Basic parameters

| Parameter | Value |
|---|---|
| Length | 312 m |
| Diameter | hull 80 m; habitat ring 312 m |
| Mass | 10,000 tonnes |
| Capacity | 520 persons (≈200 operational crew + passengers) |
| Autonomy | unlimited (closed loop + energy from space) |

## 7.2 Hull [E/S]

A graphene–diamond composite with carbon nanotubes: ~100× stronger than steel, ~10× lighter than aluminum, with a radiation-absorbing layer and limited self-repair (nanostructures in the matrix). Double-shell construction: outer shell as a Whipple shield (micrometeoroids at 0.1c), inter-shell space (coolant, storage, repair systems), inner shell (pressure integrity, insulation, load-bearing structure).

## 7.3 Spatial layout (bow → stern)

Science Section → Habitat Section (rotating) → Command Center → Engineering Section → Reactor and Drive (Modulator at the center of mass) → Bussard Collector (stern).

## 7.4 Onboard systems [E]

- **Power.** Generation: Bussard fusion up to 10 GW, deployable solar panels 100 MW @1 AU, emergency RTGs 10 MW. Storage: SMES 100 GWh, graphene ultracapacitors 10 GWh, hydrogen

fuel cells (reserve). Distribution: high-temperature superconductors, three redundant grids, AI management.

- **Life support.** Atmosphere 78% $N_2$ / 21% $O_2$ / 1% other, 101.3 kPa; oxygen production (electrolysis + algae + gardens). Water: 1000 tonnes in closed loop, 99.99% recycling, replenishment with cometary ice. Food: 5000 $m^2$ of hydroponics, protein bioreactors, 3D printing, a two-year stock of freeze-dried supplies.

- **Artificial gravity.** Rotating habitat section: 2 rpm (full rotation every 30 s), radius 156 m – a ring with a diameter equal to the hull's length, **0.7 g**; gyroscopic stabilization; can be stopped (0 g) for experiments. *(Note: during the thrust phase – see Part II.5 – the crew is in free fall; rotation provides gravity only in the cruise phase.)*

- **Computers and AI.** Quantum computer ~10,000 qubits + classical layer ~100 PFLOPS; three independent backups.

- **Communications.** Laser link 1 Gbps @1 AU, directional antenna (reserve); experimental entanglement-based channel **[S]**.

- **Shielding.** The Scoop's magnetic field as radiation shield (crucial at 0.1c). The number-one kinetic threat is not micrometeoroids but **dust**: at 0.1c a milligram grain carries the energy of ~100 kg of TNT. Layered defense: long-range lidar on the flight axis → laser ablation (point defense) → ionization of the remnants and deflection by the Scoop's field → the double-shell Whipple shield and ablative armor as the last line. The flight axis is the most closely guarded cone of sky.

- **Exploration.** ~20 laboratories; 4 landers (up to 10 t), 10 drones, **2 submarines** (for the exploration of moons with subsurface oceans), 50 EVA suits (48 h autonomy, −200…+200 °C).

# PART VIII. NAVIGATION – "THE MAP OF FALLING" (SNAV)

## 8.1 The problem of relativistic navigation

At speeds of the order of 0.1c, classical references fail, and every collision with a macroscopic object on the "fall path" is a collision at the speed of light. Navigation must be **autonomous, continuous, and anticipatory** – relying on observables that *grow* with speed, rather than fading.

## 8.2 The aberration of starlight as signal – StarNAV/SNAV

The SNAV system (inspired by the real StarNAV architecture: autonomous navigation by the relativistic perturbation of starlight, NASA) **treats aberration not as noise but as a source of information [F]**. As the speed increases, the stars crowd toward the direction of flight, forming a "tunnel of light"; the angle of that tunnel is a logarithmic indicator of speed:

- **Aberration** – from the distribution of angles between stars in the SNAV field of view, the velocity vector is determined in the ICRS frame;

- **Doppler** – blueshift ahead and redshift astern provides an independent measurement of speed and orientation;

- **CMB** – the anisotropy of the microwave background (the kinematic dipole) as an absolute velocity reference relative to the medium.

Numerical example: at a velocity of 8.1% c the aberration reaches 4.6° and the time-dilation factor is 1.0033 – values fully sufficient for precise estimation of the state vector.

## 8.3 Navigation hardware [E]

- **Main Navigation Telescope (MNT):** reflector ~2 m, actively cooled to ~40 K, CCD array of gigapixel resolution.

- **Navigation computer (SNAV):** integrates MNT data, corrects errors, determines the velocity vector (v) and position (x) in ICRS, reading aberration as signal.

## 8.4 The "fall path" – trajectory planning

The quantum computer calculates the optimal **fall path** in advance, taking into account the galactic gravitational field, the maximum acceleration (1 g), and the energy balance – analogously to ballistics, but in four-dimensional spacetime. The return is planned "backwards": one chooses the target moment, computes the predicted position of the destination (e.g., the Sun), and designs the maneuvers so that the integrated trajectory (with relativity and the fields along the way) intersects that position; the course is continuously verified against observations of stellar aberration.

# PART IX. PERFORMANCE AND MISSION PROFILE

## 9.1 Performance [E/S]

- Acceleration: comfortable 0.1 g / standard 1 g / emergency up to 5 g. At a continuous 1 g the ship would reach 10% c in ~5 weeks – in practice the limit is not thrust but the **energy balance**: the boost is spread over years at a low average acceleration (~0.01 g), with windows of full 1 g near energy reserves. 1 g is a peak capability, not a continuous regime.

- Speed: practical **10% c** (≈30,000 km/s); theoretical ceiling ~30% c; the limiter is collisions with interstellar matter.

- Range: without refueling, ~10 light-years; with refueling at stars – unlimited.

## 9.2 Proxima Centauri mission profile (≈40 years)

1. **Launch and boost (years 0–5):** orbit → heliopause; refueling at the Sun (0.1 AU); acceleration to ~5% c.

2. **Cruise (years 5–35):** ~10% c, Modulator drive, Scoop dormant; free flight, gravity from rotation.

3. **Braking (years 30–40):** flip maneuver and deceleration at 0.1 g by the Modulator; Scoop in "upstream" mode (gradient reversal – the most energetically expensive phase).

4. **Exploration (year 40+):** entry into the target system.

## 9.3 Emergency scenarios [E]

- **Modulator failure** → the Scoop takes over propulsion; the voyage is slower (a return on the order of 200 years instead of 40), the crew in hibernation **[S]**.

- **Interruption of "falling" in flight** → the ship does not "hang": when the gradient vanishes, it returns to inertial motion with the velocity accumulated (the first law of motion) – a controlled shutdown consists in gently flattening the gradient, not in an abrupt stop.

- **Loss of plasma confinement / decompression / failure of the AI layer** → redundancy procedures (three power grids, three computational backups, double-shell hull).

# PART X. THE PATH TO REALIZATION – FROM THE LABORATORY TO AN ORBITAL SHIPYARD

## 10.1 The principle of staging

The QFHPS development program is deliberately sequential: each stage has its own falsifiable pass criterion (Part XI), and moving to the next requires not faith but a result. A substantial part of the necessary industrial base already exists: high-temperature superconductors, nanometer lithography, attosecond metrology, spherical tokamaks, and magnetic sails are the subject of ongoing research programs, not fantasies.

## 10.2 The L5 orbital shipyard

Final assembly of the ship should take place in orbit, at the L5 libration point – gravitationally stable and located outside Earth's gravity well. The condition is not aesthetic but structural: the Modulator's spherical array and the Scoop's large-scale coils cannot be assembled and tuned in a 1 g field.

## 10.3 Program milestones

1. Measurement of the net force from an asymmetric Casimir cavity (ground laboratory; the key experiment, Part XI.1).
2. MEMS resonator array – demonstration of contribution summing and scaling.
3. Low-thrust orbital demonstrator (net force of the order of newtons in microgravity).
4. Demonstrator of plasma confinement and oxygen catalysis for the Scoop.
5. Mastery of high-flux heat rejection (ultimately: the neutrino channel).
6. Lifting and assembly of the hull skeleton at L5.
7. Integration of the fusion reactor ($Q > 50$ in orbital conditions).
8. Synchronization of the full Modulator array with attosecond diagnostics (coherence $\Psi_{coh}$).
9. Trial refueling at perihelion (0.1 AU) and full integration of both subsystems.

# PART XI. EXPERIMENTAL VERIFICATION AND RESEARCH PROGRAM

An engineering hypothesis is worth as much as its falsifiability. Below is the verification program – from the laboratory bench to the drive demonstrator.

## 11.1 The key experiment: the asymmetric Casimir cavity

**Goal:** to measure the net force from an asymmetric configuration of Casimir plates – proof that modulating the boundaries of the vacuum yields a *directional* impulse.

**Apparatus (low-cost variant):** two silicon plates 15×15 mm coated with gold (~100 nm), a piezo actuator (precision 5–10 nm), a Michelson interferometer as detector (sensitivity ~λ/10), a vacuum chamber at $10^{-4}$ Torr; vibration isolation on a granite slab set on bins of sand. The experiment is deliberately within reach of a well-equipped university laboratory.

**Prediction:** an asymmetric force of the order of single piconewtons, directional and repeatable, with a sign consistent with the model (2.3); higher-type cavity geometries should yield a gain of 1–2 orders of magnitude. **Success criterion:** significance ≥ 5σ over a series of repetitions with full control of systematic effects (electrostatics, thermal gradients, patch forces). From piconewtons to the target $10^6$ N, ~18 orders of magnitude separate us – the road "from a whisper to a detonation". This experiment does not build a drive; it decides whether the road exists at all. The rest would be engineering: very hard, but engineering, not magic.

## 11.2 The program of ten experiments

| # | Experiment | What it verifies | Target status |
|---|---|---|---|
| 1 | Asymmetric Casimir cavity | net force from the vacuum | key experiment – first step |
| 2 | MEMS resonator array | summation of contributions, scaling | demonstrator |
| 3 | Phase coherence Ψ_coh | THz synchronization of many cavities | demonstrator |
| 4 | Attosecond diagnostics (plasma lens) | detection of phase errors | integration |
| 5 | Dynamic modulation (DCE) | emission of real photons from the vacuum | replication of Wilson 2011 + scale |
| 6 | Measurement of metric perturbation | h_μν signature in an interferometer | critical / [S] |
| 7 | Quantum levitron / metamaterial radiator | heat management | engineering |
| 8 | Bussard-Core-1 demonstrator | fusion Q>10, oxygen catalysis | containerized reactor |
| 9 | MHD conversion | efficiency > 70% | integration |
| 10 | Integrated propulsion demonstrator | "falling" as technology | program goal |

**The goal of the program** is not a ship straight away, but a **working prototype of a propulsion modulator** that would prove that "falling in spacetime" is not only a theory but a technology. Only such a prototype would justify the last, most expensive stage – the construction of a full-scale ship.

## 11.3 Observational predictions (falsification)

The theory of the drive generates testable predictions **[E/S]**:

1. **The gravitational signature of the drive.** An object with an active Modulator-type drive would exhibit non-gravitational accelerations without visible mass ejection – a signature resolvable by precise astrometry (acceleration with no coma and no momentum flux that could account for it).

2. **Transient vacuum anisotropy.** In the vicinity of a working Modulator, a local, directional change in the density of vacuum states is predicted – measurable interferometrically (a fringe shift dependent on orientation).

3. **Neutrino flux.** Neutrino cooling implies a detectable, isotropic flux of neutrinos with a characteristic energy spectrum, correlated with drive power.

4. **DCE emission.** THz modulation of the cavity array should generate a measurable excess of photons above the thermal background, with a spectrum consistent with the dynamical Casimir effect.

Failure of any of predictions 1–4 falsifies the corresponding element of the concept.

# PART XII. THE LIMITS OF KNOWLEDGE – WHAT WE DO NOT CLOSE

Honesty requires openly marking the boundary between extrapolation and speculation.

- **Conversion of ZPE → useful momentum at ship scale [S].** The dynamical Casimir effect is confirmed; the passage from "real photons from a modulated mirror" to a thrust of 1000 kN remains unproven. This is the *proper* limit of the concept.

- **Coherence Ψ_coh [S].** The phase-coherence operator of a million resonators has no full formalism; without it there is no macroscopic force.

- **Metric perturbation h_μν [S].** The kinship with the Alcubierre metric is explicit, but the mechanism for producing useful curvature *without* exotic matter of negative energy density remains a hypothesis.

- **Coupling to the neutrino channel [S].** The principle (neutrino cooling) is hard astrophysical physics; its controlled induction in a device is not.

Here the specification ends and humility begins. These limits are not the study's shame – they are its road map.

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## CLOSING NOTE

The ship this specification concerns does not push its way through the Universe. It **falls** into it – into a valley it carves for itself, carried by the same tendency that makes a stone roll down a slope and a star exhale its heat in the form of neutrinos. The key to the stars turned out not to be a new force that had to be invented. It turned out to be an old truth of the Universe that had to be understood – and that we had to dare to trust.

No element marked [F] in this study is waiting for new physics. The Casimir force has been measured. The dynamical Casimir effect has been confirmed in the laboratory. Every neutron star cools by neutrino emission. Classical novae burn hydrogen in the hot CNO cycle. The magnetic sail has been worked out since 1991. What is missing is not a miracle but vision and will: people, institutions, and states ready to walk the program of Part XI step by step – from two gold-plated plates on a granite slab to a shipyard at the L5 point.

A humanity deprived of a common goal disperses into particular interests and loses its way. A great goal is not a luxury of a civilization – it is its salvation and its contribution to the future. Why reaching for the stars is something more than expansion, the author develops in a related cosmological work (*Cosmic Theory of Consciousness and Universe Inheritance – a scientific synthesis*, DOI: 10.5281/zenodo.23244167; Polish original: 10.5281/zenodo.23220961).

"Impossible" is usually just "not yet".

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